First prelim 2026: Dynamic Programming

Your identifier: _____

Consider the following infinite-horizon dynamic programming problem. A certain student can be either a devil or an angel. He can choose to be mean or to be nice. The reward function is:

$$r(x,u) = \begin{cases} 5 & \text{if } (x,u) = (\text{devil, mean}) \\ 0 & \text{if } (x,u) = (\text{devil, nice}) \\ -10 & \text{if } (x,u) = (\text{angel, mean}) \\ 20 & \text{if } (x,u) = (\text{angel, nice}) \end{cases}$$

The transition rule is given by:

$$M(u) = \begin{cases} \begin{array}{|c|cc|} \hline & \text{devil tomorrow} & \text{angel tomorrow} \\ \hline \text{devil now} & 1 & 0 \\ \text{angel now} & 1/2 & 1/2 \\ \hline \end{array} & \text{if } u = \text{mean} \\ \\ \begin{array}{|c|cc|} \hline & \text{devil tomorrow} & \text{angel tomorrow} \\ \hline \text{devil now} & 0 & 1 \\ \text{angel now} & 0 & 1 \\ \hline \end{array} & \text{if } u = \text{nice} \end{cases}$$

The discount factor is $\beta = 0.8$.

Problem 1: Policies

Write down all the possible policies for this student. (Hint: there are four.)

Answer: A policy specifies an action for each state $\{\text{devil}, \text{angel}\}$. Since there are two actions in each state, there are $2^2 = 4$ policies:

- π_1: (devil $\to$ mean, angel $\to$ mean)
- π_2: (devil $\to$ mean, angel $\to$ nice)
- π_3: (devil $\to$ nice, angel $\to$ mean)
- π_4: (devil $\to$ nice, angel $\to$ nice)

Problem 2: Howard's method

Guess that the policy $\pi(x) =$ nice for both states. Write the two equations that describe the conditional value function $V_\pi(x)$.

Answer: Under this policy:

- From devil: next state is angel with probability 1
- From angel: next state is angel with probability 1

Thus the value function satisfies:

$$V_\pi(\text{devil}) = 0 + \beta V_\pi(\text{angel})$$

$$V_\pi(\text{angel}) = 20 + \beta V_\pi(\text{angel})$$

Solve the second equation:

$$V_\pi(\text{angel}) = \frac{20}{1 - 0.8} = 100$$

Then:

$$V_\pi(\text{devil}) = 0.8 \cdot 100 = 80$$

Problem 3: The Bellman equation

The actual value function is:

$$V(x) = \begin{cases} 80 & \text{if } x = \text{devil} \\ 100 & \text{if } x = \text{angel} \end{cases}$$

Write the Bellman equation for when the student is a devil.

Answer: The Bellman equation implies:

$$V(\text{devil}) = \max\left\{\underbrace{5 + 0.8V(\text{devil})}_{\text{mean}},\ \underbrace{0 + 0.8V(\text{angel})}_{\text{nice}}\right\}$$

Plug in values:

Mean:

$$5 + 0.8 \cdot 80 = 69$$

Nice:

$$0 + 0.8 \cdot 100 = 80$$

Thus:

$$V(\text{devil}) = 80$$

and the optimal choice is **nice**.

For an angel:

Mean:

$$-10 + 0.8\left(\tfrac{1}{2}80 + \tfrac{1}{2}100\right) = -10 + 72 = 62$$

Nice:

$$20 + 0.8 \cdot 100 = 100$$

Thus:

$$V(\text{angel}) = 100$$

and the optimal choice is **nice**.

Hence the optimal policy is:

$$\pi^*(x) = \text{nice for both states}$$

Answer:

$$V(\text{devil}) = \max\{5 + 0.8V(\text{devil}),\ 0 + 0.8V(\text{angel})\}$$

Problem 4: The student is impatient

Solve for the value function if $\beta = 0$.

Answer: When $\beta = 0$, the student only cares about current rewards:

$$V(x) = \max_u r(x, u)$$

For a devil:

$$\max\{5, 0\} = 5$$

For an angel:

$$\max\{-10, 20\} = 20$$

Thus:

$$V(x) = \begin{cases} 5 & \text{if } x = \text{devil} \\ 20 & \text{if } x = \text{angel} \end{cases}$$